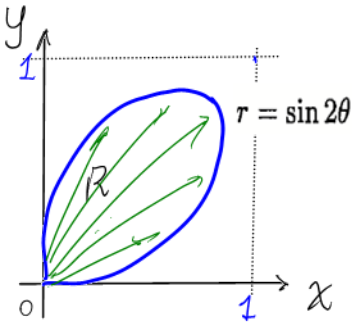
MA1521 L1 Week 08 Wed 20261007

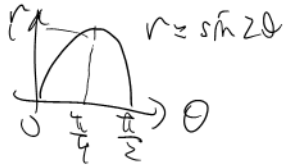
Polar coordinates : $x = r\cos\theta$, $y = r\sin\theta$, $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}\frac{y}{x}$, $\underbrace{dx\,dy}_{dA} = r\,dr\,d\theta$

Evaluate the double integral

$$\iint_R \frac{dA}{r}; \qquad R \text{ is the region inside the first-quadrant loop of } r = \sin 2\theta.$$



$0 \le \theta \le \frac{\pi}{2}$

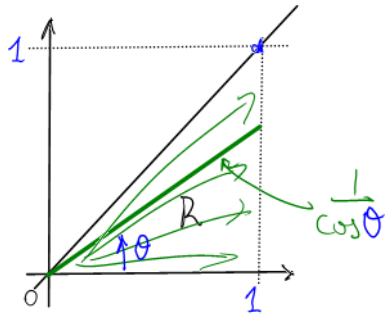


$$\iint_R \frac{dA}{r} = \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{r=0}^{r=\sin 2\theta} \frac{1}{r}\,\underbrace{r\,dr\,d\theta}_{dA}$$

$$= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \sin 2\theta\,d\theta = \left[-\frac{1}{2}\cos 2\theta\right]_{\theta=0}^{\theta=\frac{\pi}{2}} = 1$$

Evaluate the double integral

$$\iint_R \tan^2\theta \, dA; \qquad R \text{ is the triangle with vertices at } (0,0),\ (1,0),\ (1,1).$$



$$\iint_R \tan^2\theta \, dA = \int_{\theta=0}^{\theta=\frac{\pi}{4}} \int_{r=0}^{r=\frac{1}{\cos\theta}} \tan^2\theta \cdot \underbrace{r \, dr \, d\theta}_{dA}$$

$$= \int_{\theta=0}^{\theta=\frac{\pi}{4}} \tan^2\theta \left(\int_{r=0}^{r=\frac{1}{\cos\theta}} r \, dr \right) d\theta$$

$$\left. \frac{1}{2} r^2 \right|_{r=0}^{r=\sec\theta}$$

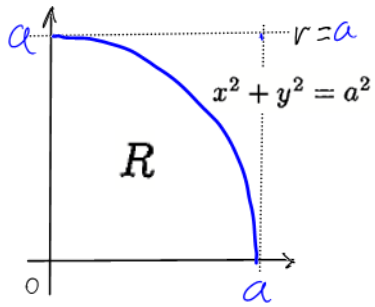
$$= \int_{\theta=0}^{\theta=\frac{\pi}{4}} \frac{1}{2} \tan^2\theta \sec^2\theta \, d\theta = \left[\frac{1}{6} \tan^3\theta \right]_{\theta=0}^{\theta=\frac{\pi}{4}}$$

$$= \frac{1}{6}$$

Evaluate the double integral $(a \in \mathbb{R}_{>0})$

$$\iint_R \frac{dx\,dy}{1+x^2+y^2};$$

R is the first-quadrant portion of the interior of $x^2+y^2=a^2$



$$\iint_R \frac{dx\,dy}{1+x^2+y^2} = \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{r=0}^{r=a} \underbrace{\frac{1}{1+x^2+y^2}}_{\frac{1}{1+r^2}} \underbrace{dx\,dy}_{r\,dr\,d\theta}$$

$$= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \left(\int_{r=0}^{r=a} \frac{r}{1+r^2}\,dr \right) d\theta$$

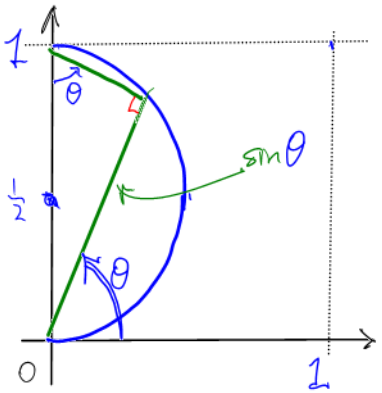
$$= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \left[\tfrac{1}{2}\ln(1+r^2) \right]_{r=0}^{r=a} d\theta$$

$$= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \tfrac{1}{2}\ln(1+a^2)\,d\theta = \tfrac{\pi}{4}\ln(1+a^2)$$

Evaluate the double integral

$$\iint_R \frac{dx\,dy}{\sqrt{1-x^2-y^2}};$$

R is the right half-disk of radius $\frac{1}{2}$ centered at $(0, \frac{1}{2})$.



$$\iint_R \frac{dx\,dy}{\sqrt{1-x^2-y^2}} = \int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{r=0}^{r=\sin\theta} \frac{1}{\sqrt{1-x^2-y^2}} \, dx\,dy$$

$$= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \left(\int_{r=0}^{r=\sin\theta} \frac{1}{\sqrt{1-r^2}} \, r \, dr \right) d\theta$$

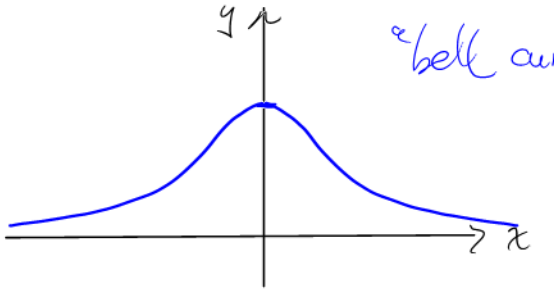
$$= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \left[-\sqrt{1-r^2} \right]_{r=0}^{r=\sin\theta} d\theta$$

$$= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \left(1 - \cos\theta \right) d\theta = \frac{\pi}{2} - \left[\sin\theta \right]_{\theta=0}^{\theta=\frac{\pi}{2}}$$

$$= \frac{\pi}{2} - 1$$

#

Key application (of polar coords)



"bell curve"

Evaluate $I = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\, dx$ (Gaussian integral)

Idea: evaluate I^2 instead:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\, dx\right)\left(\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}}\, dy\right) = \int_{y=-\infty}^{y=\infty}\int_{x=-\infty}^{x=\infty} e^{-\frac{(x^2+y^2)}{2}}\, dx\, dy$$

whole xy-plane

(use polar coords)

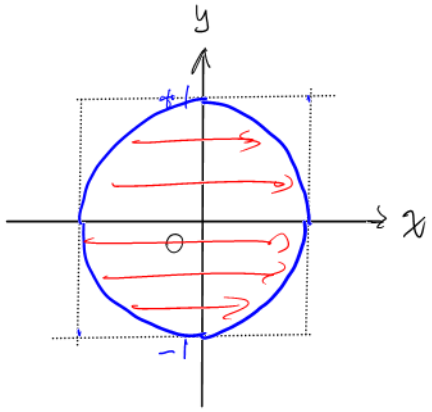
$$= \int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=\infty} \underbrace{e^{-\frac{r^2}{2}}}_{e^{-\frac{(x^2+y^2)}{2}}}\ \underbrace{r\, dr\, d\theta}_{dx\, dy}$$

$$= \int_{\theta=0}^{\theta=2\pi} \left[-e^{-\frac{r^2}{2}}\right]_{r=0}^{r=\infty} d\theta = \int_{\theta=0}^{\theta=2\pi} 1\, d\theta = 2\pi$$

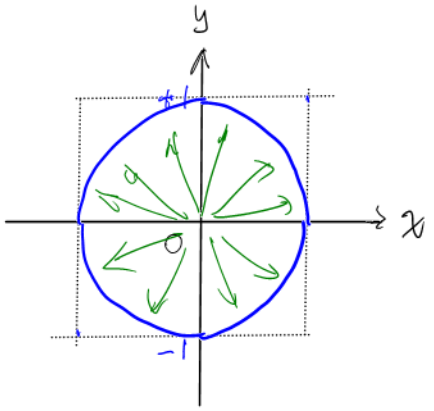
Therefore, $I = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\, dx$ is $\sqrt{2\pi}$

Evaluate $\displaystyle\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \ln(x^2+y^2+1)\,dx\,dy$

bdry satisfies $x^2 = 1-y^2$



$$\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \ln(x^2+y^2+1)\,dx\,dy$$



$$= \int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=1} \underbrace{\ln(1+r^2)}_{\ln(x^2+y^2+1)}\ \underbrace{r\,dr\,d\theta}_{dx\,dy}$$

use polar coords

$$= 2\pi \int_{r=0}^{r=1} \ln(1+r^2)\, r\,dr$$

$\int$ by parts

we subst. $u = 1+r^2$, $du = 2r\,dr$, $\Rightarrow r\,dr = \frac{1}{2}du$

$$= 2\pi \int_{u=1}^{u=2} \tfrac{1}{2}\ln u\,du = \pi\,[u\ln u - 1]_{u=1}^{u=2} = \pi\cdot(2\ln 2 - 1)$$